Ruth Cuaron

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PROFESSIONAL PROFILE

Hands-on Quantum Technician student with direct laboratory experience in optical systems, vacuum technology, and quantum instrumentation. Brings a unique foundation in precision electronics manufacturing — including soldering and board-level testing — combined with strong technical problem-solving skills developed across multiple industries.

EDUCATION

Quantum Technician Certification Program
Central New Mexico Community College - Albuquerque, NM – Spring 2026

IPC J-STD-001 Certification
Central New Mexico Community College - Albuquerque, NM – Spring 2026

Certificates: Bookkeeping, Accounting & Payroll
Central New Mexico Community College - Albuquerque, NM – 2022

SKILLS

Quantum, Optics & Photonics
- Quantum hardware & sensing: superposition, entanglement, measurement, Rabi flopping
- Qubit platforms & tradeoffs: neutral atom, trapped ion, photonic, superconducting, spin
- Optical systems: lenses, gratings, aberrations, microscopes
- Laser & photonics: laser safety, beam characterization, interferometry, polarization, optical tweezers
- Fiber optics: single/multimode fiber, coupling/splicing, photodiodes
- Spectroscopy & Metrology: absorption/emission spectroscopy, spectrometry, interferometer, precision measurement

Vacuum, Manufacturing & Lab Systems
- High and ultra-high vacuum systems: assembly, pump/gauge operation, helium leak detection, contamination control
- Vacuum technologies: turbomolecular, backing, cryogenic, ion pumps; Pirani & ion gauges
- Lab & manufacturing safety: LOTO, laser safety, optics, vacuum, cryogenics, electronics

EXPERIENCE

CNM Quantum Technician Boot Camp — Albuquerque, NM
Quantum Technician Trainee | *Albuquerque, NM 2026*
- Performed hands-on laboratory experiments involving lasers, optics, interferometry, and quantum technologies
- Built, assembled, and tested optical and experimental systems
- Collected, recorded, and analyzed experimental data
- Troubleshot laboratory equipment and experimental setups
- Worked collaboratively on team-based technical projects
- Presented technical concepts and project results to instructors and peers
- Utilized laboratory instrumentation and measurement equipment in a controlled lab environment

Bookkeeper & Tax Preparer *2019 – Present*
Hobby Machining LLC | Rio Rancho, NM
- Manage full-cycle bookkeeping for a sole proprietor business using QuickBooks and Microsoft Excel
- Prepare and file annual federal and state business tax returns for 4+ consecutive years

- Submit quarterly reports through the New Mexico Taxation and Revenue Department
- Maintain accurate financial records, reconcile accounts, and track business expenses year-round

Customer Service Representative *2003 – 2016*

Various Telecommunications Companies | Albuquerque, NM

- Provided customer service and technical support across multiple telecommunications companies
- Handled phone-based troubleshooting, warranty exchanges, and account management for major carriers

Production Technician & Material Handler *1998 – 2001*

Rockwell Semiconductor / Conexant Systems | El Paso, TX

- Executed precision soldering and assembly of printed circuit boards (PCBs) in a high-volume production environment
- Conducted board-level testing and quality inspection to verify components met production specifications
- Advanced to Material Handler role, coordinating inventory, packing, and movement of electronic components and assemblies
- Developed strong attention to detail and fine motor precision working with sensitive electronic hardware

ADDITIONAL SKILLS

Lab Equipment: Lasers · Beam Splitters · Optical Mounts · Vacuum Systems · Optical Tweezers

Lab Skills: Optical Alignment · Soldering · Board-Level Testing · Precision Assembly · Troubleshooting · Technical Documentation

Software & Tools: SolidWorks CAD · Bambu 3D Printing · Microsoft Excel

Professional Skills: Attention to Detail · Analytical Thinking · Procedure Development · Team Collaboration · Independent Worker

Languages: English · Spanish